



PowerPoint Presentation by
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MANAGEMENT ACCOUNTING

8th EDITION

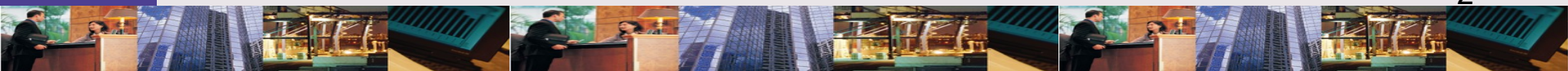
BY

HANSEN & MOWEN

15 QUALITY COSTS & PRODUCTIVITY

LEARNING OBJECTIVES

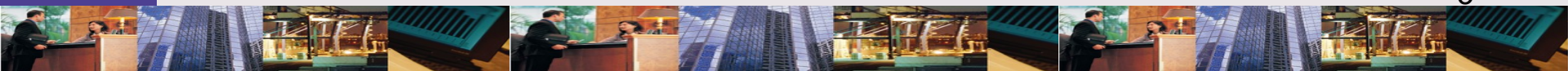
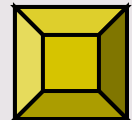
After studying this chapter, you should be able to:



LEARNING OBJECTIVES

1. Identify & describe the 4 types of quality costs.
2. Prepare a quality cost report; differentiate between acceptable quality level & total quality control.
3. Tell why quality cost information is needed & show how it is used.
4. Explain what productivity is; calculate the impact of productivity changes on profits.

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT: Ladd Lighting Corporation

Why has the measurement of
productivity & quality become
so important?



QUESTIONS TO THINK ABOUT: Ladd Lighting Corporation

What are quality costs?

QUESTIONS TO THINK ABOUT: Ladd Lighting Corporation

What kinds of quality cost reports should be prepared by the Accounting Department?

QUESTIONS TO THINK ABOUT: Ladd Lighting Corporation

What is meant by “productivity?”

LEARNING OBJECTIVE

1

Identify & describe
the 4 types of quality
costs.

QUALITY

Russell Walsh of Ladd Lighting recognizes that quality improvement can increase profitability by:

- ✧ Increasing customer demand
- ✧ Decreasing costs

WEIGHING COSTS & BENEFITS

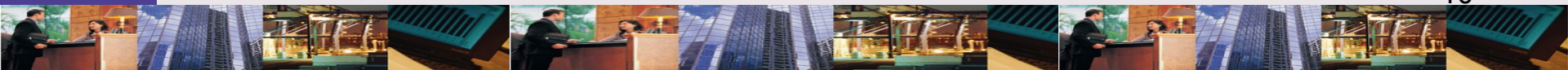
Managers need to know what quality costs are & how they change over time

- ✧ Costs of quality

- ✧ Studies suggest that cost of quality production might be as much as 20% – 30% of sales

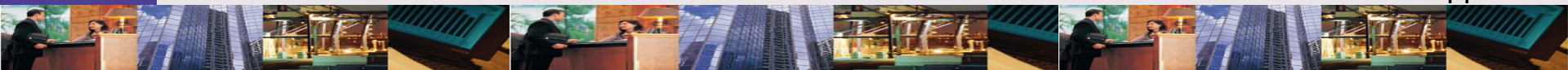
- ✧ Benefits of quality

- ✧ Competitive dimension



QUALITY PRODUCT, SERVICE: Definition

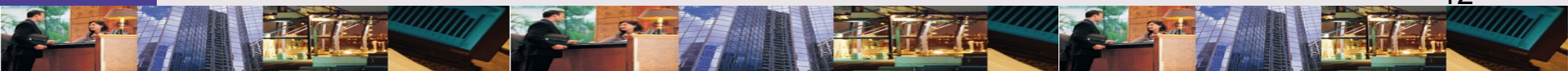
Is one that meets or exceeds
customer expectations.



DIMENSIONS OF QUALITY: 1

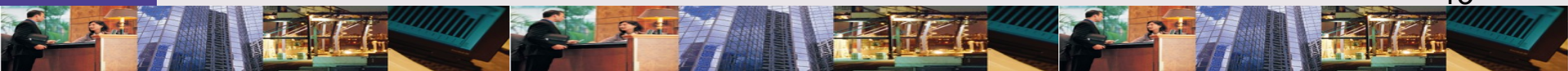
- ✳ Performance: how consistently a product functions
- ✳ Aesthetics: appearance of tangible products, facilities, communication materials
- ✳ Serviceability: ease of maintaining, repairing product
- ✳ Features of quality design: characteristics that differentiate between similar products

Continued



DIMENSIONS OF QUALITY: 2

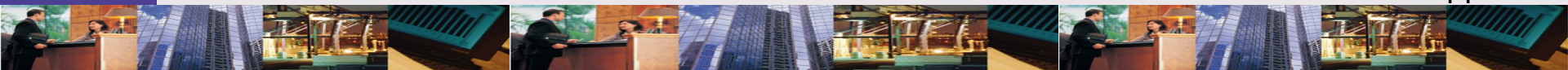
- ✧ Reliability: probability that product, service will perform intended function for specified length of time
- ✧ Durability: length of time a product functions
- ✧ Quality of conformance: measure of how a product meets its specifications
- ✧ Fitness for use: suitability of product for advertised functions



DEFECTIVE PRODUCT:

Definition

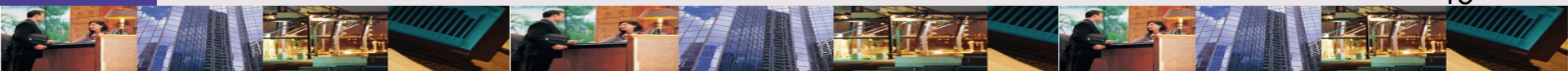
Is one that does not conform to specifications. Zero defects is the goal.



What are costs of quality?

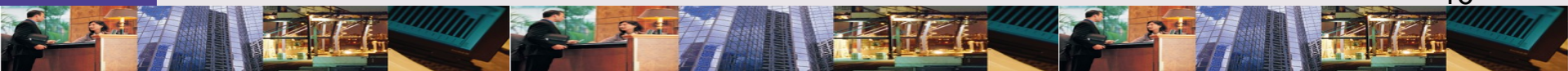
Costs that exist because poor quality does or may exist:

- **Control activities** to prevent, detect poor quality.
- **Failure activities** are responses to poor quality.



CATEGORIES OF QUALITY COSTS

1. Prevention costs: incurred to prevent poor quality
2. Appraisal costs: incurred to determine whether products, services conform to requirements, customer needs
3. Internal failure costs: incurred when non-conformance discovered & product, service re-worked, scrapped, etc.
4. External failure costs: incurred when products fail to conform after delivery and recalled



CLASSIFYING QUALITY COSTS

✧ Observable

- ✧ Costs available in accounting records

✧ Hidden

- ✧ Significant

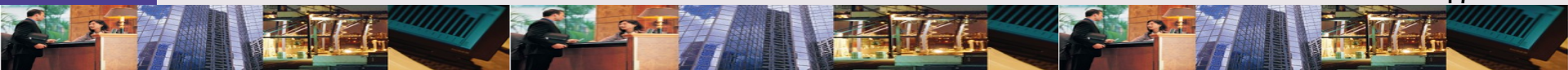
- ✧ Not directly available in accounting records

- ✧ Estimated

- ✧ Multiplier method

- ✧ Market research

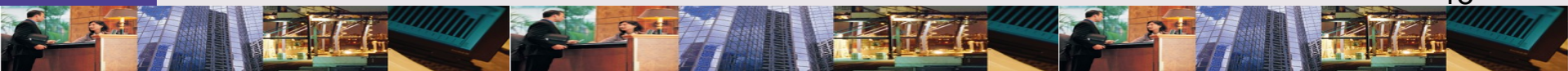
- ✧ Taguchi quality loss function



FORMULA: Multiplier Method

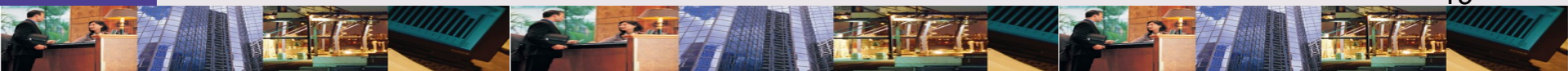
Multiplier method estimates quality costs as some multiple of measured failure costs.

Total external failure cost:
= k (Measured external failure costs)



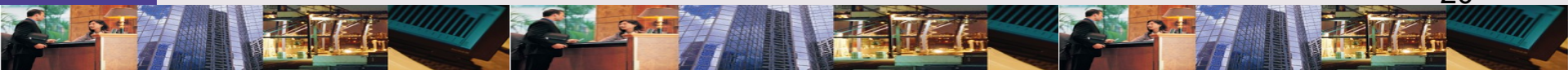
How does market research
estimate hidden quality
costs?

Market research uses customer
surveys & interviews of sales
staff to project future profit
losses.



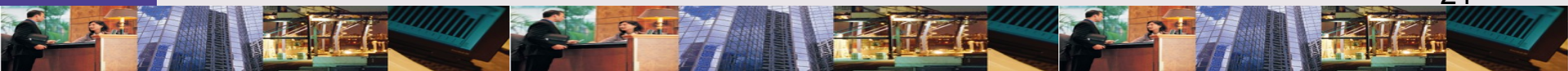
SPECIFICATION LIMIT: Definition

In traditional quality model,
defines the area of acceptable
quality around the target value.

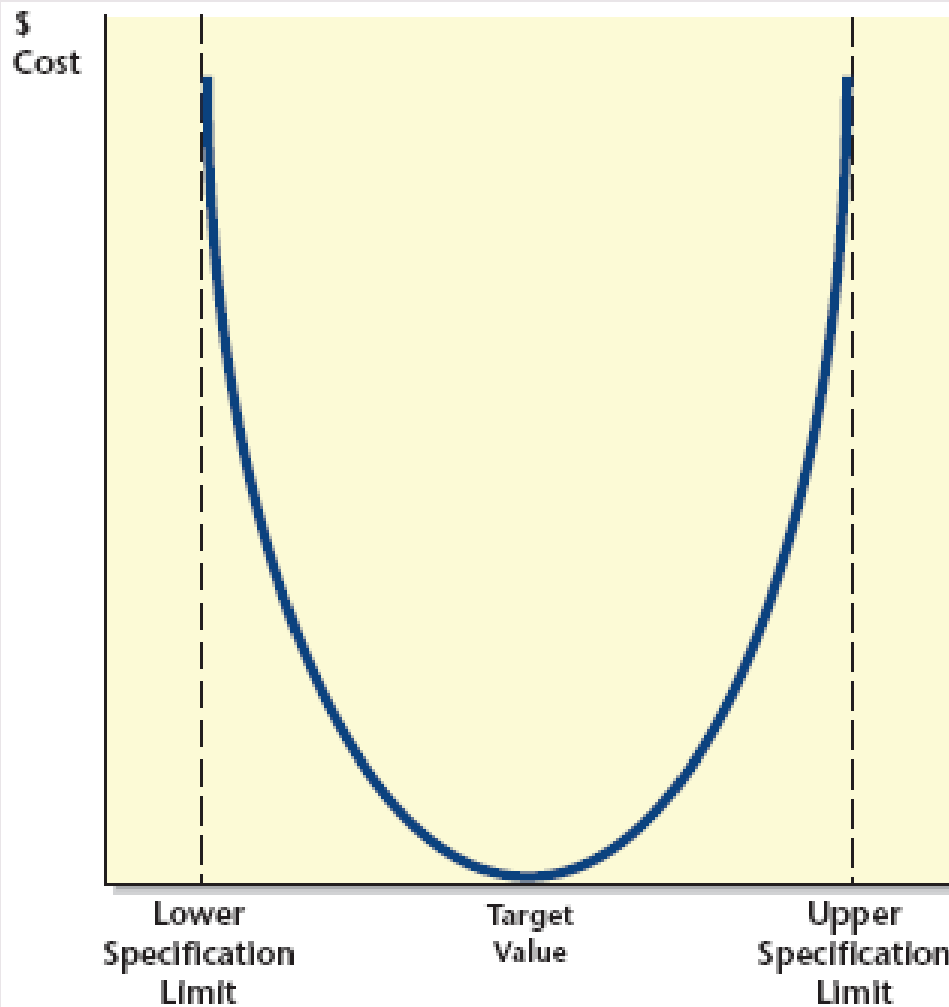


What assumption does the Taguchi quality loss function make?

Taguchi quality loss function assumes that variations from target value of quality characteristic causes hidden quality costs regardless of specification limits.



TAGUCHI QUALITY LOSS FUNCTION



Quality cost increases symmetrically at an increasing rate even within specification limits.

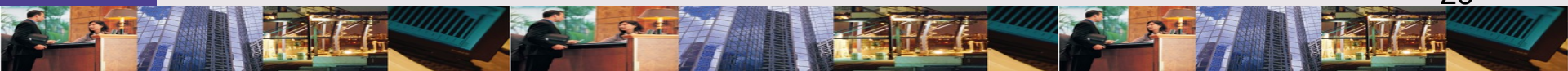
FORMULA: Taguchi Function

Taguchi quality loss function estimates hidden costs of poor quality.

[Quality loss * Actual value of quality characteristic] $L(y)$

= a proportional constant multiplier of external cost failure structure * (difference between actual and target value squared)

$$L(y) = k(y-T)^2$$



How do we estimate the organization's external failure cost structure, k ?

k is estimated as c/d^2

where:

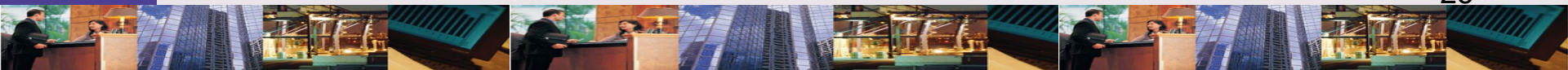
c = loss at lower or upper specification limit

d = distance of limit from target value

LEARNING OBJECTIVE

2

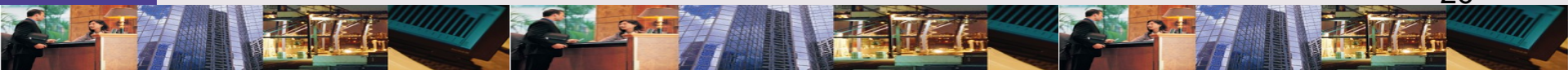
Prepare a quality cost report; differentiate between acceptable quality level & total quality control.



QUALITY COST REPORT

Provides insights to companies serious about quality:

- ✧ Reveals magnitude of quality costs by category
 - ✧ Allows managers to assess **financial impact** of quality costs in each category
- ✧ Shows distribution of quality costs by category
 - ✧ Allows managers to assess **relative importance** of each category



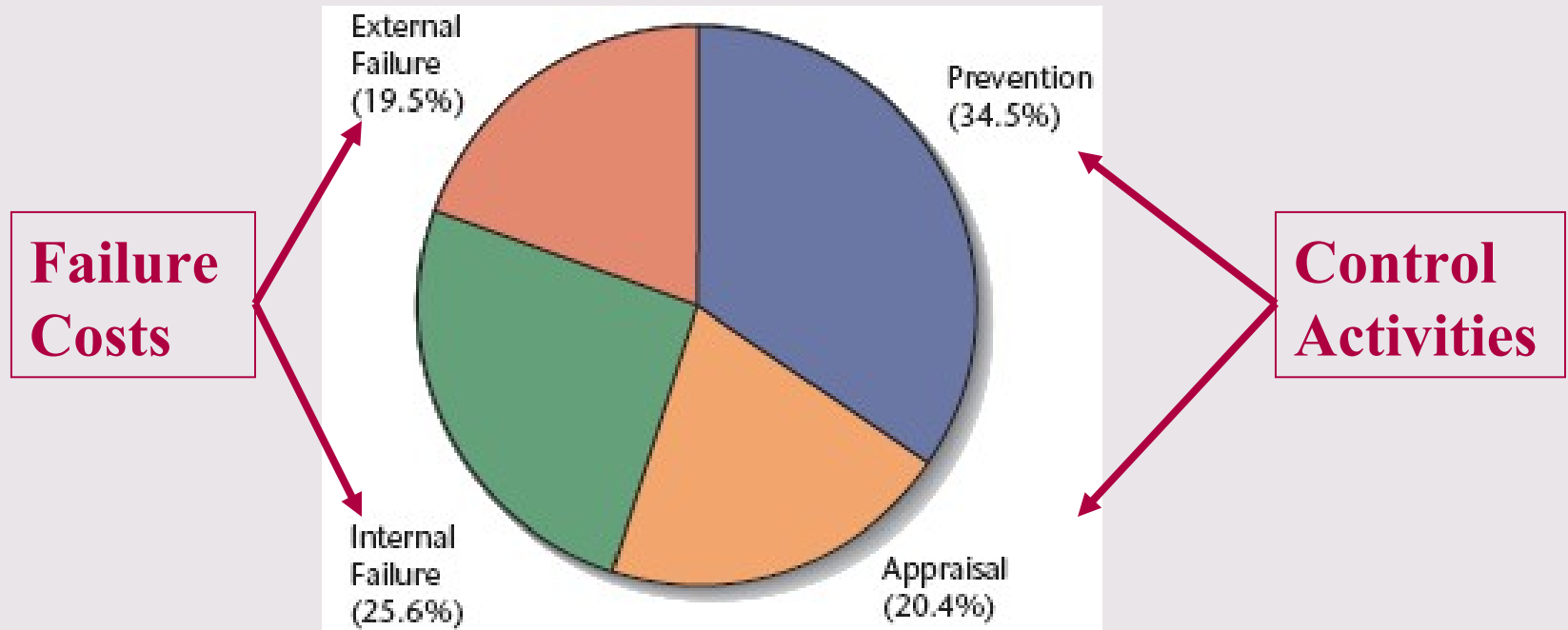
QUALITY COST REPORT

Ladd Lighting Corporation Quality Cost Report for the Year Ended March 31, 2008			
	Quality Costs		Percentage (%) of Sales ^a
Prevention costs:			
Quality training	\$350,000		
Reliability engineering	<u>800,000</u>	\$1,150,000	5.18%
Appraisal costs:			
Materials inspection	\$200,000		
Product acceptance	100,000		
Process acceptance	<u>380,000</u>	680,000	3.06
Internal failure costs:			
Scrap	\$500,000		
Rework	<u>350,000</u>	850,000	3.82
External failure costs:			
Customer complaints	\$250,000		
Warranty	250,000		
Repair	<u>150,000</u>	650,000	2.93
Total quality costs		<u>\$3,330,000</u>	<u>14.99%</u> ^b

^aActual sales of \$22,200,000.
^b\$3,330,000/\$22,200,000 = 15%; difference is rounding error.

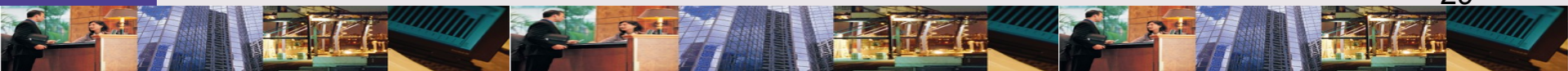
Exhibit 15-3 Quality Cost Report

QUALITY COST DISTRIBUTION



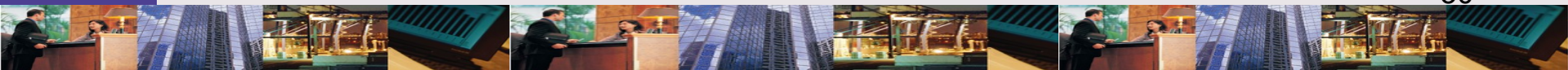
ACCEPTABLE QUALITY LEVEL (AQL): Definition

Is the **optimal balance**
between control costs &
failure costs.



Is there a problem with the
ACL (traditional) view of
quality?

AQL encouraged lower quality
levels by accepting production of
a given number of defective
units.



AQL QUALITY COST GRAPH

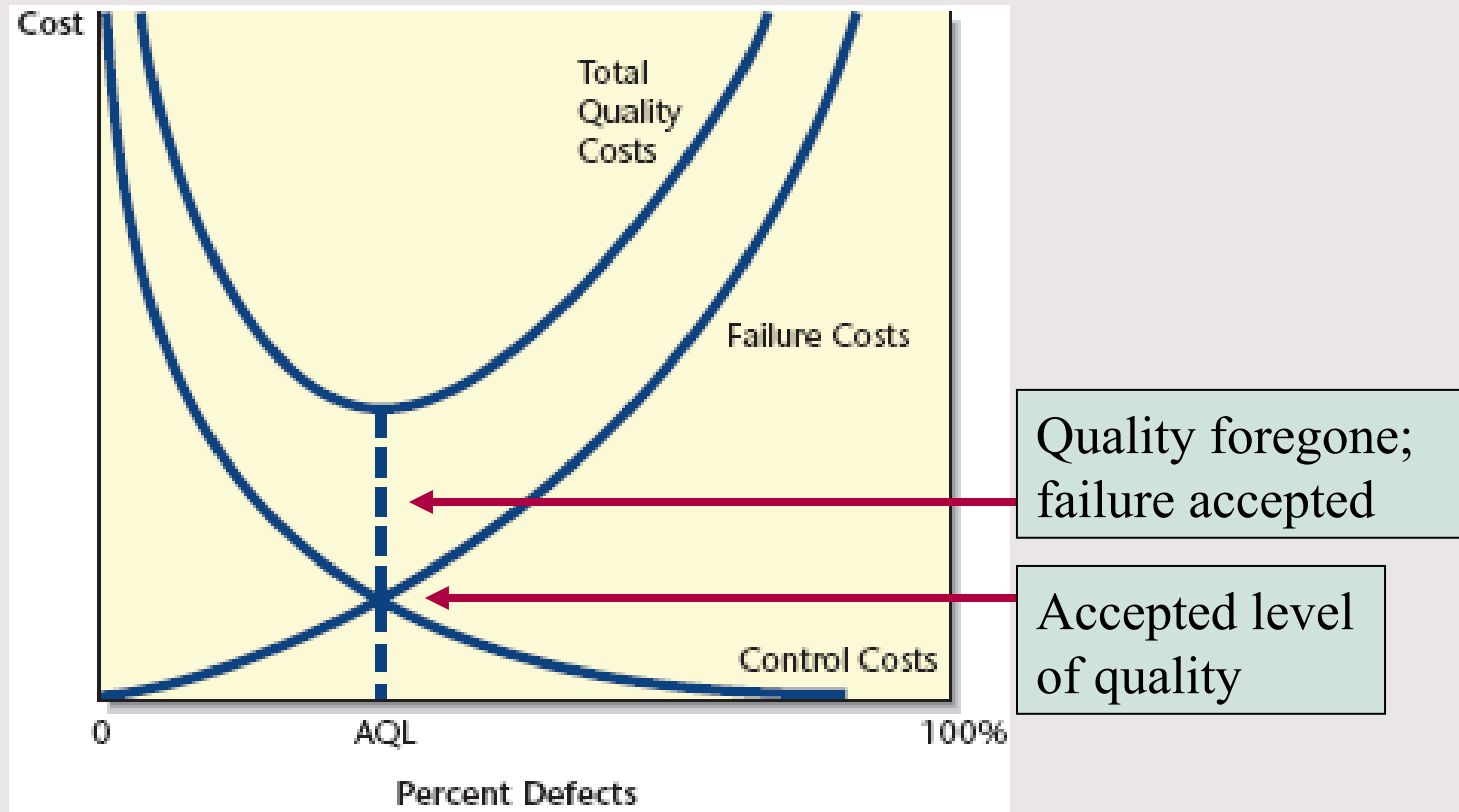
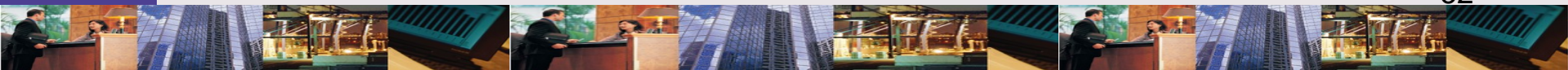


EXHIBIT 15-5

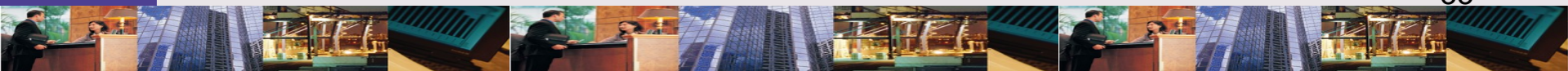
ZERO DEFECTS MODEL: Definition

Claims that it is cost
beneficial to reduce non-
conforming units to zero.

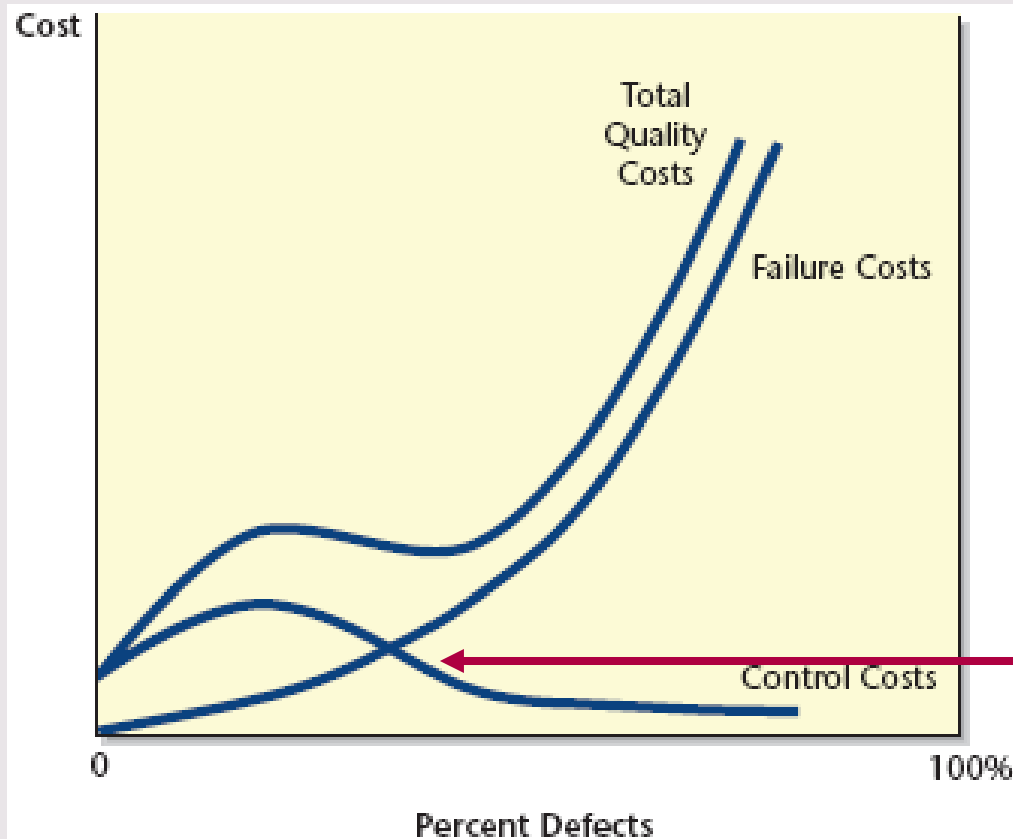


Is there a problem with the
zero defects model?

Zero defects model **understates**
quality costs & the potential for
savings from efforts to improve
quality.



AQL QUALITY COST GRAPH

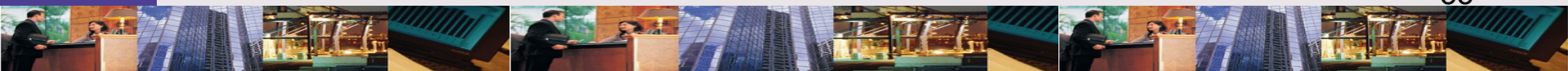


Control costs decrease as percentage of defects decreases.

EXHIBIT 15-6

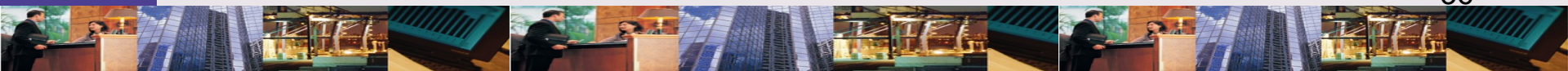
REDUCING QUALITY COSTS

- ✳ Take direct attack on failure costs to drive them to zero
- ✳ Invest in “right” prevention activities to bring about improvement
- ✳ Reduce appraisal costs according to results achieved
- ✳ Continuously evaluate, redirect prevention efforts to gain further improvement



What is the strategy for reducing costs based on?

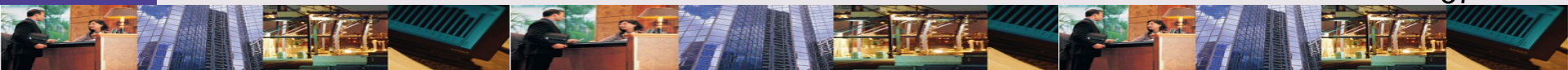
The strategy is based on the premise that a) there is a root cause for each failure, b) causes are preventable, and c) prevention is always cheaper.



ABM & OPTIMAL QUALITY COSTS

ABM classifies costs as value-added & non-value-added and recommends non-value-added costs be eliminated.

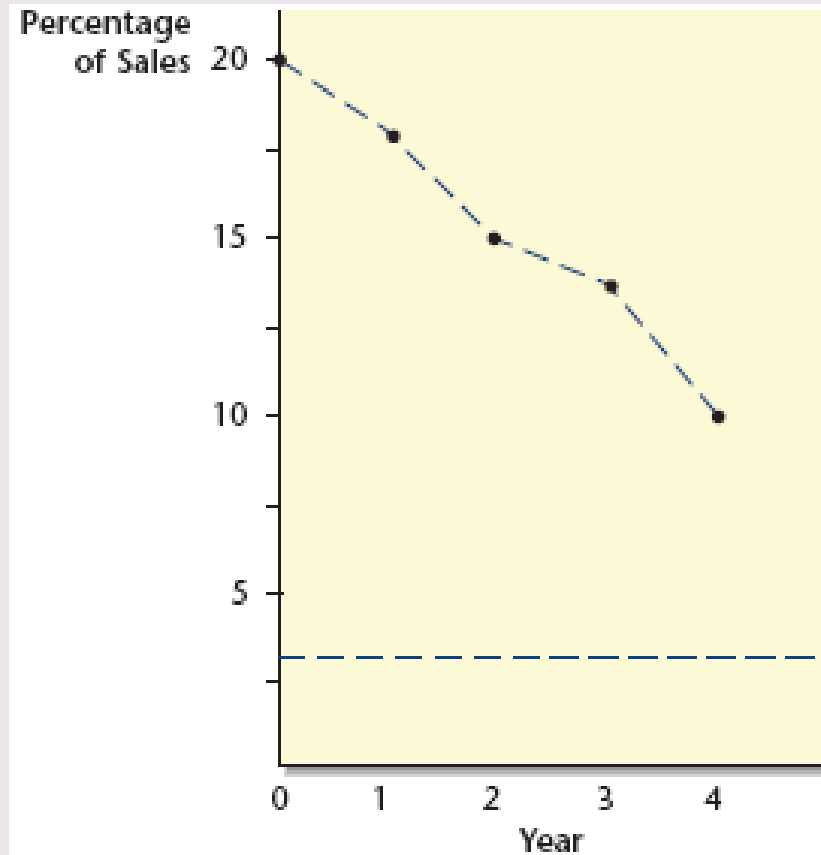
- ✧ Value-added quality costs
 - ✧ Prevention activities, when performed efficiently
- ✧ Non-value-added quality costs
 - ✧ Appraisal costs
 - ✧ Failure costs (both internal & external)



TREND ANALYSIS: TQC

	Quality Costs	Actual Sales	Costs as % of Sales
2004	\$ 440,000	\$ 2,200,000	20.0%
2005	423,000	2,350,000	18.0
2006	412,500	2,750,000	15.0
2007	392,000	2,800,000	14.0
2008	280,000	2,800,000	10.0

TQC TREND GRAPH



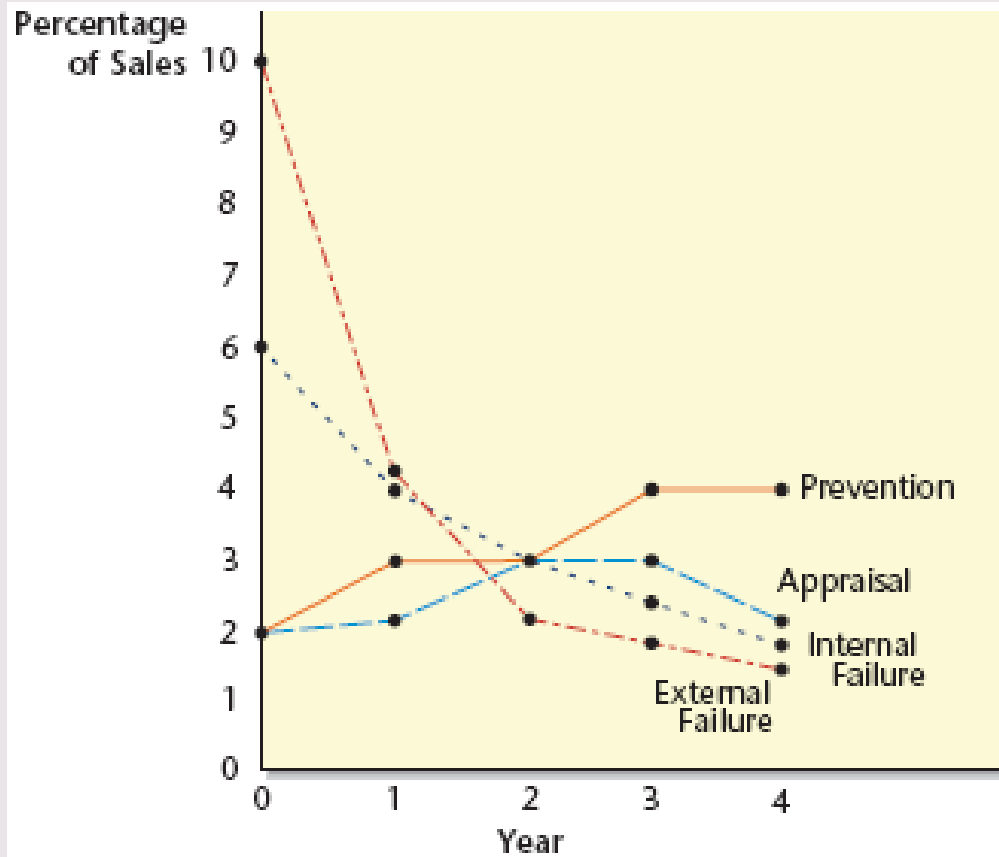
Although total quality costs are decreasing, we need to analyze its components.

TREND ANALYSIS: TQC Components

	Prevention	Appraisal	Internal Failure	External Failure
2004	2.0%	2.0%	6.0%	10.0%
2005	3.0	2.4	4.0	8.6
2006	3.0	3.0	3.0	6.0
2007	4.0	3.0	2.5	4.5
2008	4.1	2.4	2.0	1.5



TQC COMPONENT GRAPH



Over time, quality costs shift from non-value-added to value-added (prevention) costs.

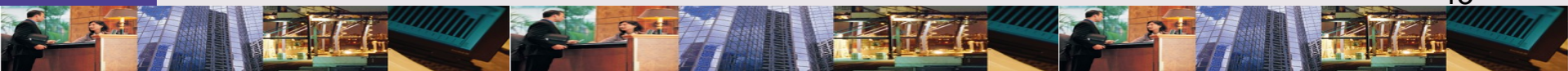
LEARNING OBJECTIVE

3

Tell why quality cost information is needed & show how it is used.

What are principal objectives of reporting quality costs?

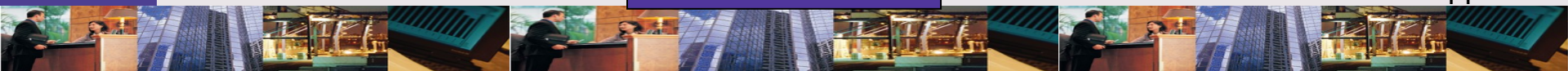
Principal objectives are to improve & facilitate a) managerial planning, b) control, and c) decision making.



STRATEGIC PRICING: Background

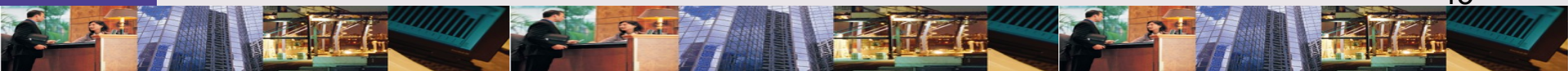
Market data for low priced electronic measurement instruments shows market share has dropped. Japanese firms continue to pressure the product line. Leola Wise is preparing a brief to support a significant (\$3) price decrease to hold or recapture market share. Quality cost estimates follow.

Continued



QUALITY COSTS: Background

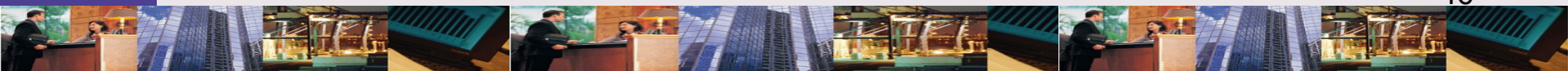
Inspection of raw materials	\$ 200,000
Scrap	800,000
Rejects	500,000
Rework	400,000
Product inspection	300,000
Warranty work	<u>1,000,000</u>
Total estimate	<u>\$ 3,200,000</u>



ELECTRONIC INSTRUMENTS:

Price Reduction Analysis

The price reduction can be achieved by a combination of implementing a total quality control position, working to reduce the cost of lower level instruments, while redesigning the production process.

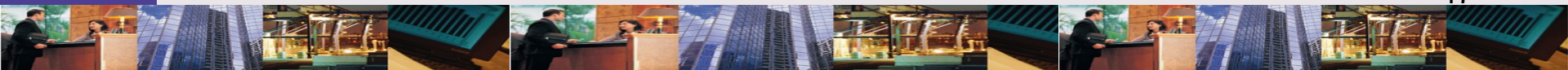


NEW PRODUCT ANALYSIS:

Background

A marketing manager and design engineer developed a proposal for a new product. They were surprised when approval was not forthcoming because the product did not meet the company-required 18% return on sales. They received a report from the controller's office with the following life-cycle profit estimates.

Continued



PROJECTED LIFE-CYCLE INCOME STATEMENT: Background

Sales (50,000 * \$60)	\$ 3,000,000
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Cost of inputs:

Materials	800,000
-----------	---------

Labor	400,000
-------	---------

Scrap	150,000
-------	---------

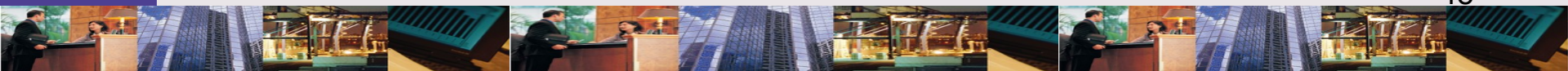
Inspection	350,000
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Repair work	200,000
-------------	---------

Product development	500,000
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Selling	<u>300,000</u>
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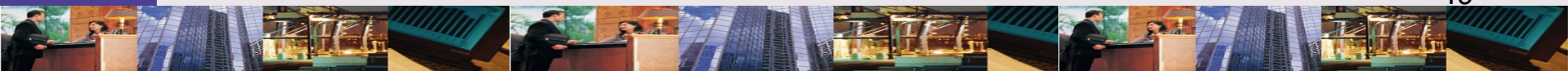
Life-cycle income	\$ <u>300,000</u>
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NEW PRODUCT: Life-Cycle Profit Analysis

A new product design would eliminate scrap and rework, leading to cost savings. Cost reductions included \$150,000 for scrap, \$200,000 for scrap, and eliminating 1 inspector at \$50,000. The new analysis suggests that the return on sales would be 30% and the new product should be accepted.

Continued



PROJECTED LIFE-CYCLE INCOME STATEMENT: Analysis

Sales (50,000 * \$60)	\$ 3,000,000
-----------------------	--------------

Cost of inputs:

Materials	800,000
-----------	---------

Labor	400,000
-------	---------

Scrap	0
--------------	----------

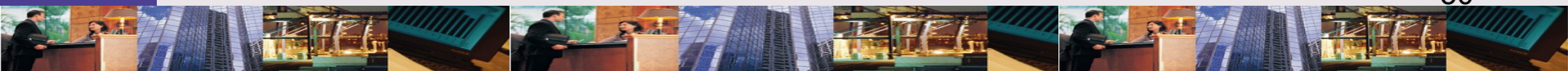
Inspection	300,000
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Repair work	0
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Product development	500,000
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Selling	<u>300,000</u>
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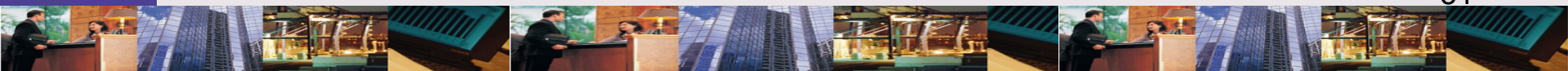
Life-cycle income	\$ <u>650,000</u>
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LEARNING OBJECTIVE

4

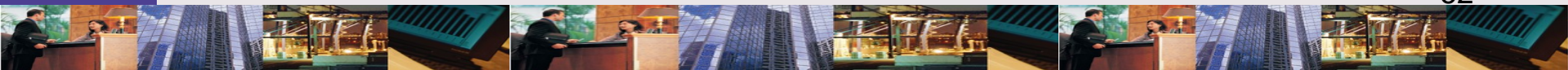
Explain what productivity is; calculate the impact of productivity changes on profits.



TOTAL PRODUCTIVE EFFICIENCY

When concerned with productive efficiency, 2 conditions must be satisfied:

- ✦ **Technical efficiency:** For any mix of inputs that will produce a given output, no more of any 1 input is used than necessary to produce the output
- ✦ **Input trade-off efficiency:** Given the mixes that satisfy the first condition, the least costly mix is chosen.



TECHNICAL EFFICIENCY IMPROVEMENTS: Panel A

Current Productivity

Inputs:

Labor 

Capital 

Output:



Same Output, Fewer Inputs

Inputs:

Labor 

Capital 

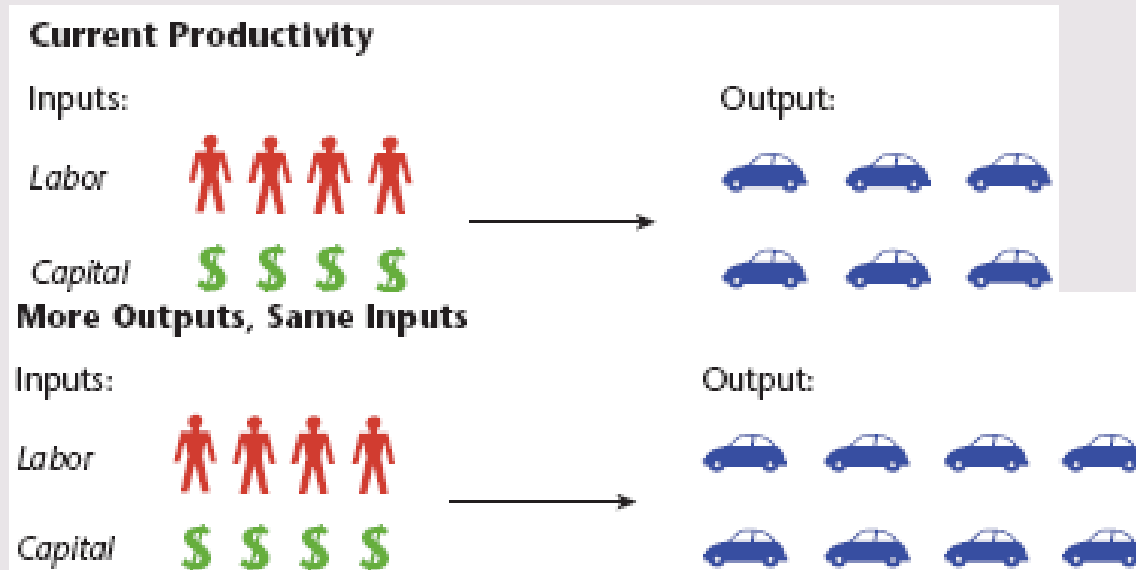
Output:



The first approach is to produce the same output with fewer inputs.

EXHIBIT 15-9

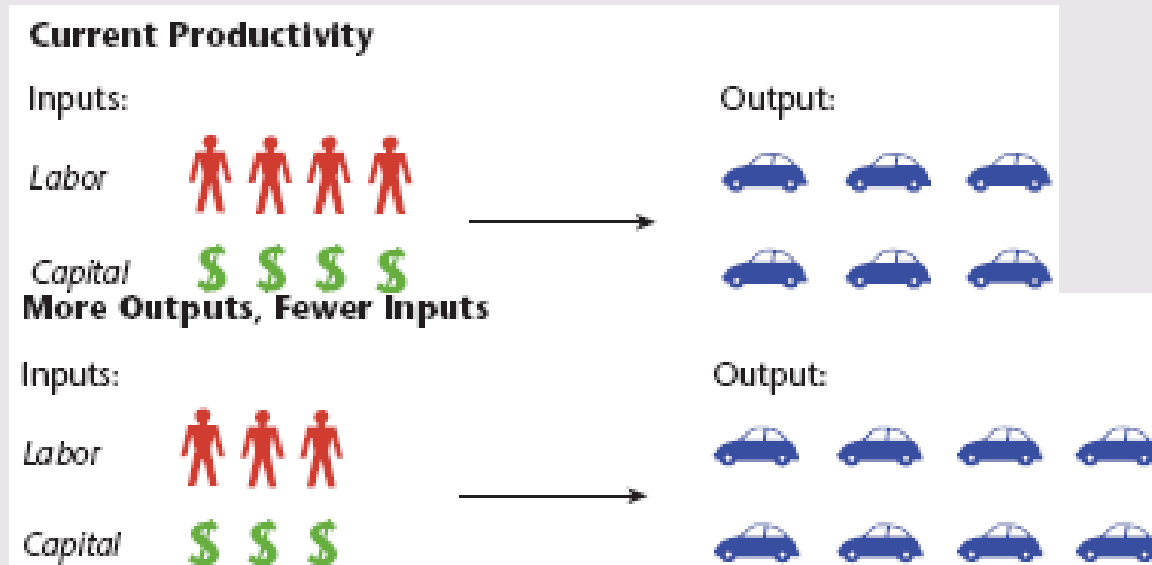
TECHNICAL EFFICIENCY IMPROVEMENTS: Panel B



The second approach is to produce more output with the same inputs.

EXHIBIT 15-9

TECHNICAL EFFICIENCY IMPROVEMENTS: Panel C

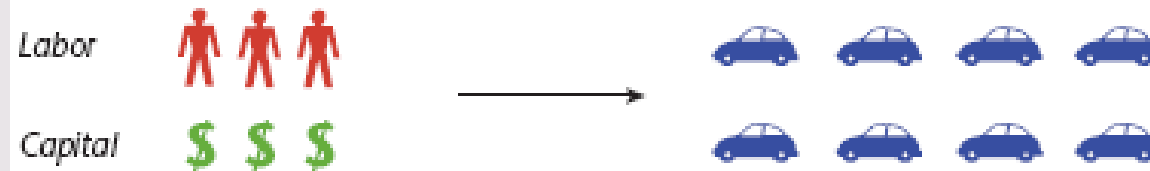


The third approach is to produce more output with fewer inputs.

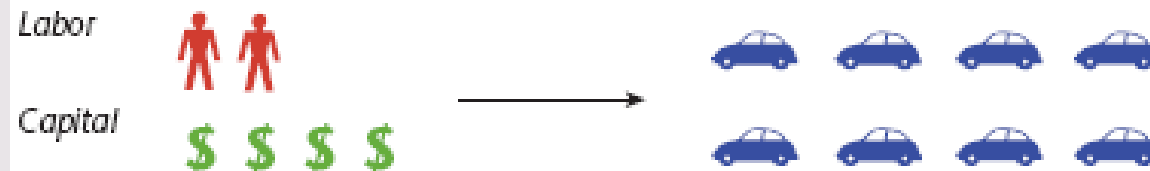
EXHIBIT 15-9

INPUT TRADE-OFF EFFICIENCY

Technically Efficient Combination I:
Total Cost of Inputs = \$20,000,000



Technically Efficient Combination II:
Total Cost of Inputs = \$25,000,000

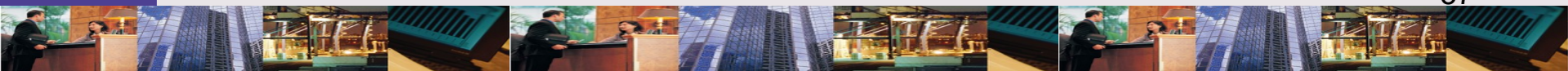


Managers must weigh the trade-off between labor & capital for efficiency of output.

EXHIBIT 15-10

PRODUCT DATA: Background

	2007	2008
# Chandeliers produced	120,000	150,000
Labor hours used	40,000	37,500
Materials used (lbs.)	1,200,000	1,428,571



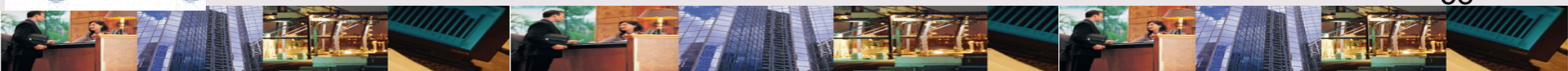
FORMULA: Partial Productivity Measurement

Partial productivity measurement is a quantitative assessment of productivity changes.

Productivity ratio = Output / Input

Operational productivity = 120,000 / 40,000
= 3 chandeliers per hour

Financial productivity = \$6,000,000 / 480,000
= \$12.50 in revenue per #1 labor cost



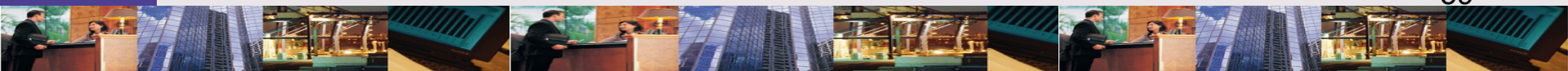
ADVANTAGES & DISADVANTAGES: Partial Measures

✧ Advantages

- ✧ Managers can focus on a particular input
- ✧ Easily interpreted
- ✧ Feedback for operational personnel

✧ Disadvantages

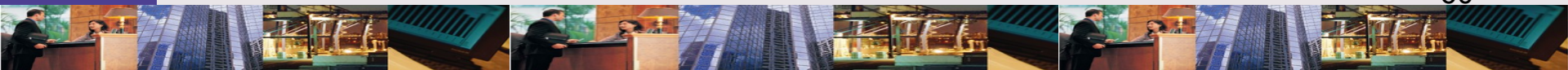
- ✧ In isolation, can be misleading
- ✧ Partial measures are not suitable for trade-offs



PARTIAL MEASURES: Analysis

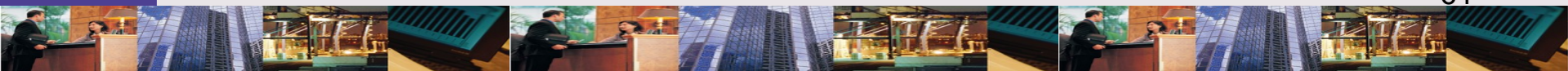
Conclusions that can be drawn about partial measures:

- ✧ Existence of trade-offs mandates total measure of productivity for assessing merits of productivity decisions
- ✧ Because of possibility of trade-offs, financial productivity must be measured



TOTAL PRODUCTIVITY MEASUREMENT: Definition

Is measuring productivity for
all inputs simultaneously.



PRODUCT DATA: Background

REPEAT

	2007	2008
# Chandeliers produced	120,000	150,000
Labor hours used	40,000	37,500
Materials used (lbs.)	1,200,000	1,428,571

PROFILE ANALYSIS: No Trade-offs

Partial productivity based on product data.

	Partial Productivity Ratios	
	2007 Profile ^a	2008 Profile ^b
Labor productivity ratio	3.000	4.000
Materials productivity ratio	0.100	0.105
^a Labor: 120,000/40,000; materials: 120,000/1,200,000		
^b Labor: 150,000/37,500; materials: 150,000/1,428,571		

Exhibit 15-11 Productivity Measurement: Profile Analysis with No Trade-offs—Ladd Lighting Corporation

PROFILE ANALYSIS: With Trade-offs

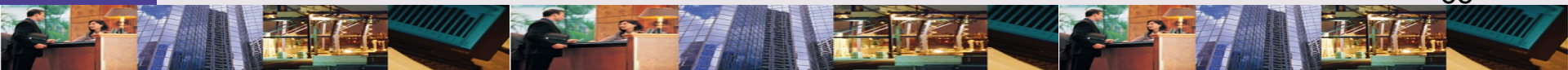
Trade-offs between inputs lowers the materials productivity ratio.

	Productivity Profile	
	2007 ^a	2008 ^b
Labor productivity ratio	3.000	4.000
Materials productivity ratio	0.100	0.088
^a Labor: 120,000/40,000; materials: 120,000/1,200,000		
^b Labor: 150,000/37,500; materials: 150,000/1,700,000		

Exhibit 15-12 Productivity Measurement: Profile Analysis with Trade-offs—Ladd Lighting Corporation

PROFIT-LINKED PRODUCTIVITY MEASUREMENT: Definition

Is measuring the amount of profit change attributable to productivity change.

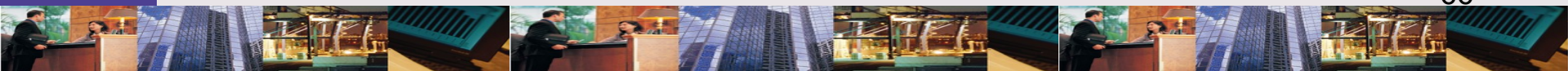


PROFIT-LINKAGE RULE:

Definition

States that productivity change is the difference between

[Cost of inputs without productivity change — cost of inputs actually used].



PRICE RECOVERY COMPONENT: Background

	2008	2007	Difference
Revenues	\$ 7,200,000	\$ 6,000,000	\$ 1,200,000
Less: Cost of inputs	<u>5,550,000</u>	<u>2,840,000</u>	<u>2,710,000</u>
Profit	<u>\$ 1,650,000</u>	<u>\$ 3,160,000</u>	<u>\$ <1,510,000></u>

FORMULA: Profit Recovery

Profit recovery is the change in revenue minus a change in the cost of inputs .

Profit recovery

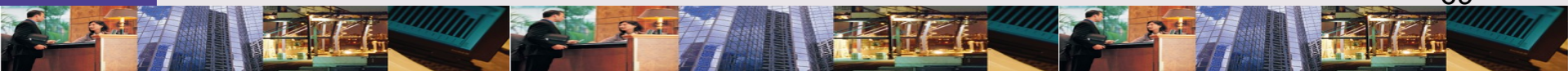
= Profit change – Profit linked productivity change

= (\$1,510,000 – \$450,000)

= \$1,060,000

GAINSHARING: Definition

Is providing to a company's entire workforce cash incentives that are keyed to quality & productivity gains



CHAPTER 15

THE END